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Tutorial Sheet

Searching and Sorting

Q.1) Let a be an array containing n integers in increasing order. The following algorithm determines (searches) whether there are two distinct numbers in the array whose difference is a specified number S>0.

```
i = 0; j = 1;
while(j < n){
        if(E) j++;
        else if ( a[j] – a[i] == S) break;
        else i++;
}
If (j<n) printf("Yes");
else printf("No");
```

Choose the correct expression for E.

a) a[j] – a[i] > S
b) a[j] – a[i] < S
c) a[i] – a[j] < S
d) a[i] – a[j] > S

Q.2) An array A of length n with distinct elements is said to be bitonic if there is an index 1 <= i <= n such that A[1.. i] is sorted in the non-decreasing order and A[i + 1..n] is sorted in the non-increasing order.

Which ONE of the following represents the best possible asymptotic bound for the worst-case number of comparisons by an algorithm that searches for an element in a bitonic array A?

a) θ(n)
b) θ(1)
c) θ($\log^2 n$)
d) θ(log n)

Q.3) Consider the following C program that attempts to locate an element x in an array Y[] using binary search. The program is erroneous.

```
1.  f(int Y[10], int x) {
2.    int i, j, k;
3.    i = 0; j = 9;
4.    do {
5.          k =  (i + j) /2;
6.          if( Y[k] < x)  i = k; else j = k;
7.       } while(Y[k] != x && i < j);
8.    if(Y[k] == x) printf ("x is in the array ") ;
9.    else printf (" x is not in the array ") ;
10. }
```

On which of the following contents of Y and x does the program fail?

a) Y is [1 2 3 4 5 6 7 8 9 10] and x < 10
b) Y is [1 3 5 7 9 11 13 15 17 19] and x < 1
c) Y is [2 2 2 2 2 2 2 2 2 2] and x > 2
d) Y is [2 4 6 8 10 12 14 16 18 20] and 2 < x < 20 and x is even

Q.4) Let a and b be two sorted arrays containing n integers each, in non-decreasing order. Let c be a sorted array containing 2n integers obtained by merging the two arrays a and b. Assuming the arrays are indexed starting from 0, consider the following four statements:

I. a[i]>= b[i] -> c[2i] >= a[i]
II. a[i]>= b[i] -> c[2i] >= b[i]
III. a[i]>= b[i] -> c[2i] <= a[i]
IV. a[i]>= b[i] -> c[2i] <= b[i]

Which of the following is TRUE

a) Only I and II
b) Only I and IV
c) Only II and III
d) Only III and IV

Q.5) The usual θ (n^2) implementation of Insertion Sort to sort an array uses linear search to identify the position where an element is to be inserted we use binary search to identify the position, the into the already sorted part of the array. If instead worst case running time will.

a) remain $\theta(n^2)$
b) become $\theta(n(\log^2 n))$
c) become θ(n log n)
d) become θ(n)

Q.6) Which one the following inplace sorting algorithms needs the minimum number of swaps?

a) Quicksort
b) Insertion sort
c) Selection sort
d) Heap sort

Q.7) Consider an array $A = [10,7,8,19,41,35,25,31]$. Suppose the merge sort algorithm is executed on array A to sort it in increasing order. The merge sort algorithm will carry out a total of 7 merge operations.

A merge operation on sorted left array L and sorted right array R is said to be void if the output of the merge operation is the elements of array L followed by the elements of array R.

The number of void merge operations among these 7 merge operations is _____. (answer in integer)

Q.8) The number of swappings needed to sort the numbers 8 , 22, 7, 9, 31, 5, 13 in ascending order using bubble sort is_____

Q.9) Randomized quicksort is an extension of quicksort where the pivot is chosen randomly. What is the worst case complexity of sorting n numbers using randomized quicksort?

a) O(n)
b) O(n log n)
c) $O(n^2)$
d) O(n!)

Q.10) If one uses straight two-way merge sort algorithm to sort the following elements in ascending order:
20, 47, 15, 8, 9, 4, 40, 30, 12, 17
then the order of these elements after second pass of the algorithm is:

a) 8, 9, 15, 20, 47, 4, 12, 17, 30, 40
b) 8, 15, 20, 47, 4, 9, 30, 40, 12, 17
c) 15, 20, 47, 4, 8, 9, 12, 30, 40, 17
d) 4, 8, 9, 15, 20, 47, 12, 17, 30, 40

Answer key:

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	d	c	c	a	c	3	10	c	b